

PRIMARY NEUROENDOCRINE CARCINOMA OF THE BREAST:
A MULTIDISCIPLINARY DIAGNOSTIC AND THERAPEUTIC CHALLENGE

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1 INTRODUCTION / WHY THIS CASE MATTERS

- NECB is a rare malignancy: <1% of all breast cancers.
- Recognized distinct entity since WHO 2019.
- Overlapping features with common invasive cancers (Diagnostic challenge).
- Crucial role of imaging for detection, staging; and pathology correlation.

2 CASE AT A GLANCE

- 54-year-old female.
- Gradually enlarging right breast lump (3 months)
- 10kg weight loss.
- No pain, nipple discharge, skin changes.
- Sister diagnosed with breast cancer (age 40).
- Firm, non-tender 3cm mass in upper right breast. No axillary lymphadenopathy.

3 IMAGING FINDINGS

MAMMOGRAPHY

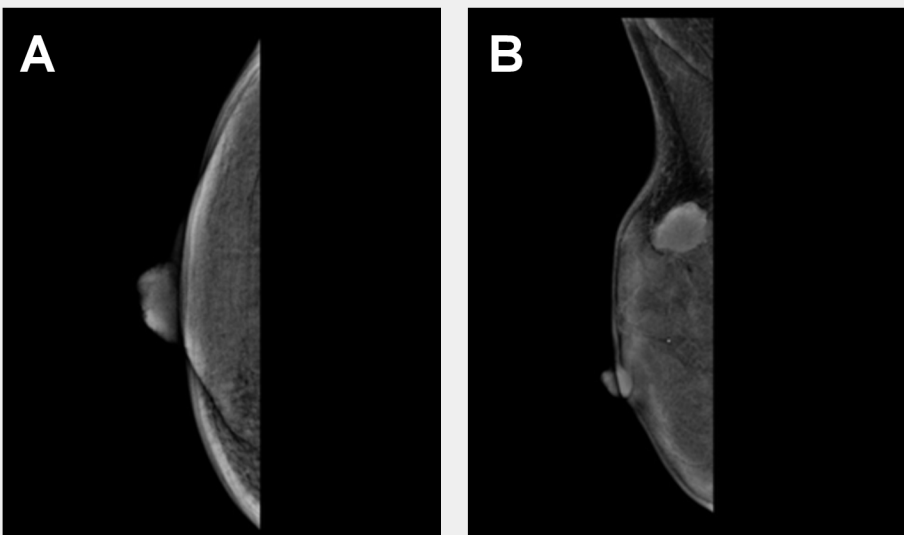


Figure 1 | Right Mammography

A | Right CC, B | Right MLO views

- Ill-defined hyperdense mass · Best seen on right MLO
- No suspicious calcification · BI-RADS 5.

PEARL: A palpable malignancy may be inconspicuous on one projection – always correlate orthogonal views and targeted ultrasound.

TARGETED ULTRASOUND

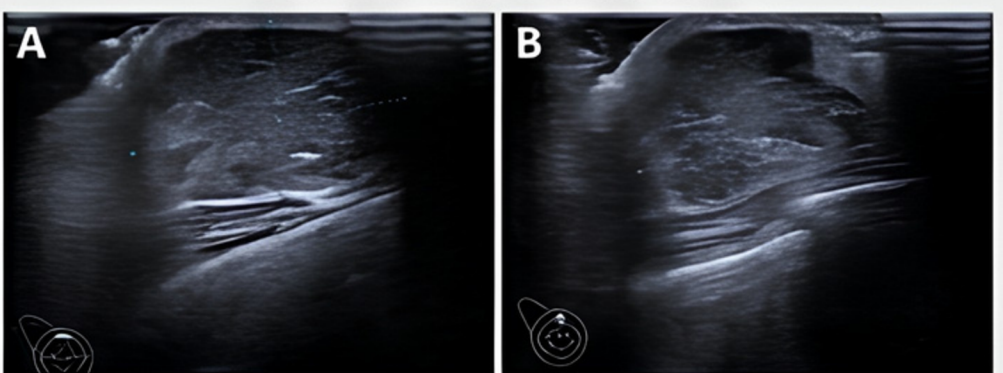


Figure 2 | Targeted Ultrasound

A | Axial gray-scale, B | Longitudinal gray-scale

An irregular hypoechoic mass with indistinct margins and heterogeneous internal echotexture

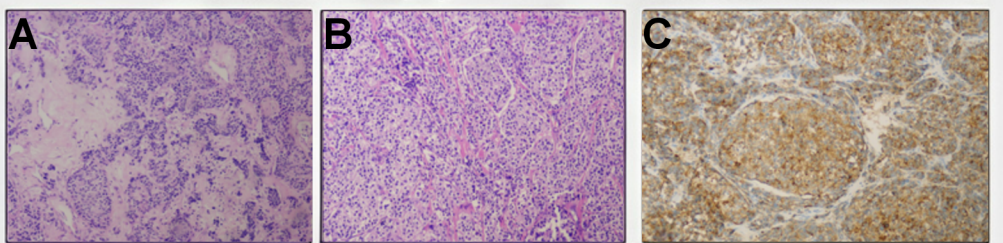
Measurement: 25 x 16 x 23 mm.

No posterior acoustic shadowing.

4 HISTOPATHOLOGY & IMMUNOHISTOCHEMISTRY

Core Needle Biopsy

Immunohistochemistry



A | H&E Low Power B | H&E High Power C | Synaptophysin Positivity (Diffuse)

- Synaptophysin: diffuse +
- ER: +
- Ki-67: ~40%
- HER2: IHC 1+ (negative)

STAGING PET-CT

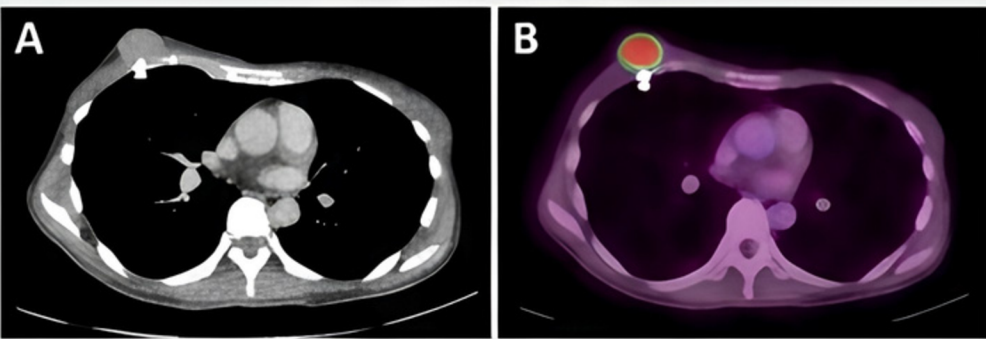


Figure 3 | Staging PET-CT

A | Contrast-enhanced CT, B | Fused FDG PET-CT.

Intense FDG uptake in breast tumor.

Confirmed localized disease; no distant metastases.

5 TREATMENT TIMELINE & OUTCOME



- Diagnosis (BI-RADS 5) → Core Biopsy
- Right Modified Radical Mastectomy + Sentinel LN Biopsy (Confirmed 3.2cm Grade 3 NECB, Associated DCIS, Positive LN)
- Completion Axillary LN Dissection
- Adjuvant Chemotherapy (Platinum-etoposide)
- Hormonal Therapy (Tamoxifen → Letrozole)
- Radiotherapy (40 Gy / 15 fractions)
- Stable condition, no evidence of recurrence.



6 RADIOLOGY DISCUSSION & KEY REPORTING TIPS



1. NON-SPECIFIC PHENOTYPE
NECB may mimic conventional invasive breast carcinoma.



2. ONE VIEW CAN MISS IT
Dense parenchyma reduced conspicuity on the CC projection.



3. MORPHOLOGY DRIVES SUSPICION
Irregular margins and suspicious morphology remain important even without posterior acoustic shadowing.



4. TISSUE DEFINES THE SUBTYPE
Neuroendocrine differentiation requires histopathology and immunohistochemistry.



5. STAGING MATTERS
PET-CT established disease extent; negative axillary imaging did not exclude microscopic sentinel-node metastasis.

7 CONCLUSIONS & TAKE-HOME MESSAGE

NECB has no pathognomonic imaging appearance and may mimic conventional invasive breast carcinoma. The radiologist's role is to recognise malignant morphology, assess disease extent, guide tissue diagnosis and correlate imaging with pathology—not to predict histological subtype from imaging alone.

8 REFERENCES

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3. Lavigne M, Menet E, Tille JC, et al. *Mod Pathol*. 2021;34(1):95-106.
4. Inno A, Bogina G, Turazza M, et al. *Oncologist*. 2016;21(1):28-32.